

RAK7289V2/RAK7289CV2 WisGate Edge Pro V2 Datasheet

Overview

Description

The RAK7289V2 WisGate Edge Pro is the flagship product of the RAK Edge Series. Featuring industrial-grade components, it ensures a high level of reliability.

It supports up to 16 LoRa channels and offers multi-backhaul options with Ethernet, Wi-Fi, and cellular connectivity. Additionally, it provides a dedicated port for various power options, including solar panels and batteries. Its redesigned enclosure accommodates LTE, Wi-Fi, and GPS antennas internally for enhanced aesthetics and functionality.

In addition, RAK7289V2 operates under [WisGateOS2](#) , which is built on the latest OpenWrt kernel. The OS Web UI features a new design and supports multiple extension installations, enabling remote management using [WisDM](#)  for personalized gateway customization.

WARNING

This product is intended to be powered by 12 V_{DC} through a designated power port. The use of solar chargers is not recommended, as they may supply overvoltage, potentially damaging the device. We strongly discourage the use of such chargers with this product, and any damage incurred as a result will void the warranty.

Features

Hardware

- IP67/NEMA-6 industrial-grade enclosure with cable glands
- PoE (802.3af) + Surge Protection

- Dual LoRa Concentrators for up to 16 channels
- Backhaul: Wi-Fi, Ethernet, LTE (optional, available with RAK7289CV2)
- GPS
- Supports 12 V_{DC} or solar power supply with electricity monitoring (Solar Kit optional)
- Internal antennas for Wi-Fi, GPS, and LTE
- External antenna for LoRa
- Dying-gasp (optional)

Software

- [WisGateOS2](#) 
- [WisGateOS2 Extensions](#)  : OpenVPN, Wireguard VPN, and others
- Remote management with [WisDM](#)  Fleet Managment
- Built-in Network Server (full LoRaWAN support v1.0.3)
- Full LoRaWAN Stack support with Semtech SX1303
- LoRa Frame filtering (node whitelisting in Packet Forwarder mode)
- MQTT v3.1 bridging with TLS encryption
- Fine timestamping (optional)
- Buffering of LoRa frames in Packet Forwarder mode in case of NS outage (no data loss)
- Listen Before Talk (optional)

Specifications

Overview

The overview presents the block diagram for the RAK7289V2/RAK7289CV2 that shows the internal architecture of the hardware.

Block Diagram

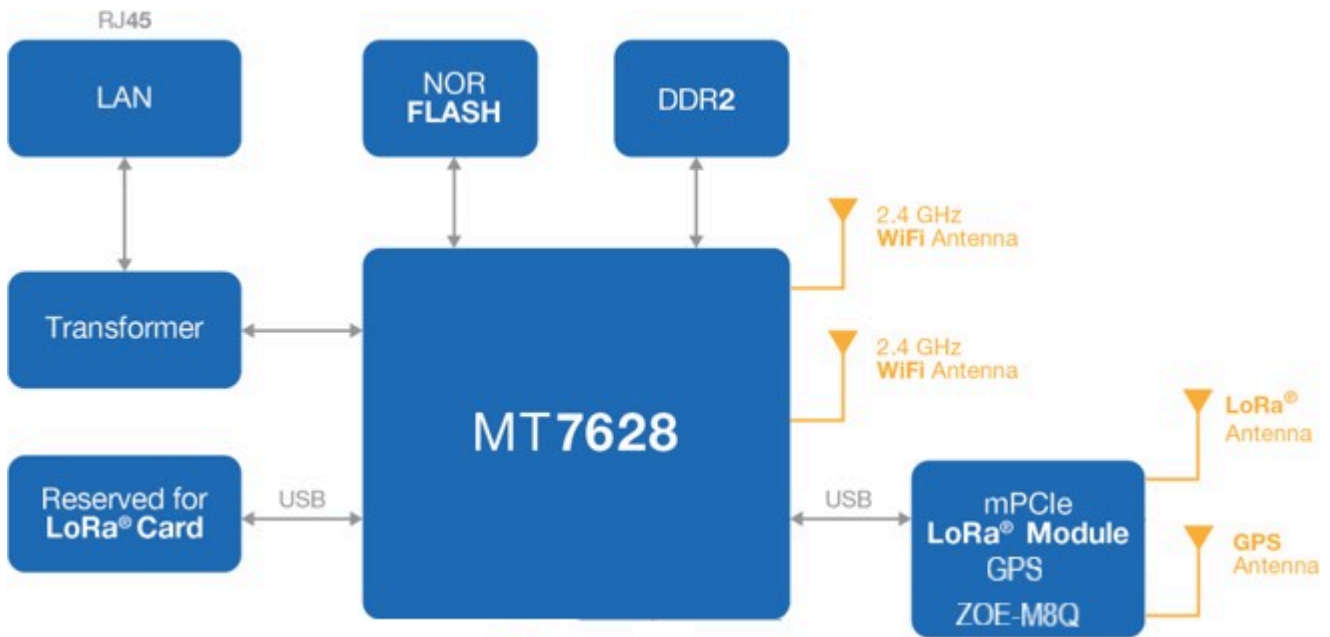


Figure 1: RAK7289V2 Block Diagram

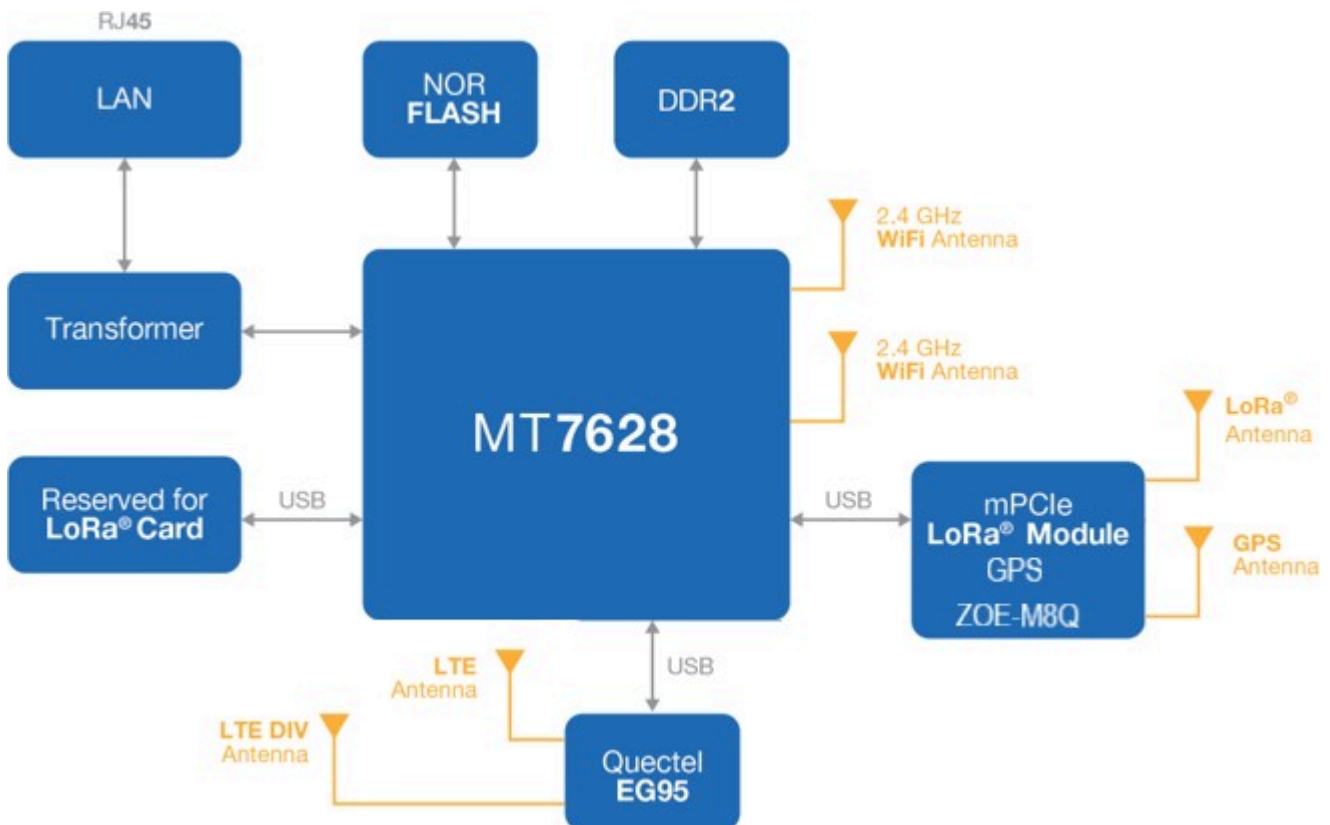


Figure 2: RAK7289CV2 Block Diagram

Main Specifications

Feature	Specifications
Computing	MT7628, DDR2 RAM 128 MB
Wi-Fi Features	Frequency: 2.4 GHz (802.11b / g / n)
	2×2 MIMO
	RX Sensitivity: - 95 dBm (Min)
	TX Power: 20 dBm (Max)
	Operation channels: 2.4 GHz: 1-13
LoRa Features	SX1303 mPCIe card (connects maximum of two)
	8 Channels (16 channels optional)
	RX Sensitivity: - 139 dBm (Min)
	TX Power: 17 dBm (Max)
	Listen Before Talk
Frequency	• EU868 • IN865 • US915 • AU915 • KR920 • AS923-1/2/3/4 • EU433 • CN470

Feature	Specifications
Cellular Features (available with RAK7289CV2)	Nano SIM Card: 12.30 mm x 8.80 mm x 0.67 mm Supports Quectel EG95-E / EG95-NA / EC25-J / EC25-AU(IoT / M2M -optimized LTE Cat 4 Module)
	EG95-E for EMEA Region (Europe, Middle East and Africa)
	LTE FDD: B1/B3/B7/B8/B20/B28A
	WCDMA: B1/B8
	GSM/EDGE: B3/B8
	EG95-NA for North America Region
	LTE FDD: B2/B4/B5/B12/B13
	WCDMA: B2/B4/B5
	EC25-J for Japan Region
	LTE FDD: B1 / B3 / B8 / B18 / B19 / B26
	LTE TDD: B41
	WCDMA: B1 / B6 / B8 / B19
	EC25-AU for Brazil Region
	LTE-FDD: B1 / B2 / B3 / B4 / B5 / B7 / B8 / B28
	LTE-TDD: B40

Feature	Specifications
	WCDMA: B1 / B2 / B5 / B8
	GSM/EDGE: B2 / B3 / B5 / B8
Power Supply	PoE (IEEE 802.3af) - 37~57 V _{DC} 12 V _{DC} Compatible with RAK Battery Plus
ETH	RJ45 (10/100 M)
Antenna	LoRa: N-Type connector (one for the 8-channel gateway and two for the 16-channel gateway)
	Wi-Fi / GPS / LTE: Internal antenna
Ingress protection	IP67
Enclosure material	Aluminium and plastic
Dimensions	240 mm x 240 mm x 89.5 mm
Weight	4.6 kg Gateway only
Operating temperature	- 30° C to + 55° C
Storage Temperature	- 40° C to + 85° C
Operating humidity	0% to 95% (non-condensing)
Storage Humidity	0% to 95% (non-condensing)
Installation Method	Pole or wall mounting

Hardware

The hardware specification details the interfacing of the RAK7289V2/RAK7289CV2 and its corresponding functionalities. It also outlines the parameters and standard values of the gateway.

Interfaces



Figure 3: RAK7289V2/RAK7289CV2 Interfaces

NOTE

The SD card in the SD card slot must not be ejected, as it may affect the device's performance, as various logs and data are stored on it.

Reset Key Functions

The functions of the Reset key are as follows:

- **Short press:** Restart the gateway.
- **Long press** (5 seconds and above): Restore factory settings.

LED Indicators

LEDs	Status Indication Description
LED 1 (PWR)	Power indicator: The LED is on when device power is on

LEDs	Status Indication Description
LED 2 (ETH)	ON: Link is up
	OFF: Link is down
	Flicker: Data transmitting and receiving
LED 3 (LoRa 1)	ON: LoRa 1 is working
	OFF: LoRa 1 is not working
	Flicker: Indicate LoRa 1 Packet receiving and sending
LED 4 (WLAN)	AP Mode: • ON: The AP is up • Flicker: Data receiving and sending
	STA Mode: • Slow flicker (1 Hz): Disconnected • ON: Connected • Flicker: Data receiving and sending
LED 5 (LTE)	Slow Flicker (1800 ms bright / 200 ms dark): Network searching
	Slow flicker (200 ms dark / 1800 ms bright): Idle
	Fast flicker (125 ms bright / 125 ms dark): Ongoing data transfer

LEDs	Status Indication Description
LED 6 (LoRa 2 for 16 channel)	ON: LoRa 2 is working
	OFF: LoRa 2 is not working
	Flicker: Indicate LoRa 2 Packet receiving and sending

RF Specifications

Wi-Fi Radio Specifications

Feature	Specifications
Wireless Standard	IEEE 802.11b / g / n
Operating Frequency	ISM band: 2.412~2.472 GHz
Operation Channels	2.4 GHz: 1-13
Transmit Power (The max power maybe different depending on local regulations) - per chain	802.11b
	19 dBm @1 Mbps
	19 dBm @11 Mbps
	802.11g
	18 dBm @6 Mbps
	16 dBm @54 Mbps
	802.11n (2.4 GHz)

Feature	Specifications
	18 dBm @MCS0 (HT20)
	16 dBm @MCS7 (HT20)
	17 dBm @MCS0 (HT40)
	15 dBm @MCS7 (HT40)
Receiver Sensitivity (Typical)	802.11b
	-95 dBm @1 Mbps
	-88 dBm @11 Mbps
	802.11g
	-90 dBm @6 Mbps
	-75 dBm @54 Mbps
	802.11n (2.4 GHz)
	-89 dBm @MCS0 (HT20)
	-72 dBm @MCS7 (HT20)
	-86 dBm @MCS0 (HT40)

Feature	Specifications
	-68 dBm @MCS7 (HT40)

LoRa Radio Specifications

Feature	Specifications
Operating Frequency	• EU868 • IN865 • US915 • AU915 • KR920 • AS923-1/2/3/4 • EU433 • CN470
Transmit Power	27 dBm (Max)
Receiver Sensitivity	-139 dBm (Min)

Electrical Characteristics

The gateway supports multiple power supply options.

- **Power Cord + PoE Adapter:** The gateway is powered via PoE. PoE (IEEE 802.3af), 37~57 V_{DC}.
- **DC Cable:** The gateway is powered by a customized external DC power supply. You need to use the DC cable to connect an external power supply. The external power supply voltage: 12 V_{DC}.
- The gateway also supports [RAK9155 Battery Plus](#) as its power supply. RAK9155 Battery Plus need to be purchased separately.

Mechanical Characteristics

Parameter	Value
Dimensions	240 mm x 240 mm x 89.5 mm Gateway only (no antenna, no bracket)
Weight	4.6 kg Gateway only
Ingress protection	IP67

Environmental Requirements

Parameter	Value
Operating Conditions	Operating Temperature: - 30° C to + 55° C Storage Temperature: - 40° C to + 85° C Operating Humidity: 0 ~ 95% RH non-condensing Storage Humidity: 0 ~ 95% RH non-condensing

Software

LoRa	Network	Management
Gateway OTA management	Wi-Fi AP mode	WisDM
LoRa package forward (packet forwarder, Basics Station)	Wi-Fi Client mode	SSH2, NTP
Frequency Band Setup	LTE APN Setup	Firmware update
Country Code setup	Support 802.1q	Firmware update
TX Power Setup	Uplink backup	LoRa Packet Forwarder

LoRa	Network	Management
Data logger	Firewall	Built-in Network Server
Location setup	DHCP Server/Client	MQTT Bridge
Statistic		OpenVPN, Ping Watch Dog
Supports class A, B, and C		WEB UI
Server address and Port setup		

Firmware

Model	Firmware Version	Source
RAK7289V2/RAK7289CV2 WisGate Edge Pro	v2.2.10	Download 

Models/Bundles

Part Number	8 Channel SX1303	16 Channel SX1303	Cat4 Cellular	GPS	Wi- Fi	Dying gasp
RAK7289CV2- XYZ	✓		✓	✓	✓	
RAK7289CV2- XYZ		✓	✓	✓	✓	

Part Number	8 Channel SX1303	16 Channel SX1303	Cat4 Cellular	GPS	Wi- Fi	Dying gasp
RAK7289CV2- XYZ	✓		✓	✓	✓	✓
RAK7289CV2- XYZ		✓	✓	✓	✓	✓
RAK7289V2- XYZ	✓			✓	✓	
RAK7289V2- XYZ		✓		✓	✓	
RAK7289V2- XYZ	✓			✓	✓	✓
RAK7289V2- XYZ		✓		✓	✓	✓

Certification

